

Quiz 7 tomorrow:
Solving linear systems

Today: Leslie matrices

For age-structured populations

Age 0	40% →	Age 1
Age 1	30% →	Age 2
Age 2	10% →	Age 3

Age 1	have on average 2 offspring
Age 2	" " " 1.5 "

$N_0(t)$	$N_2(t)$
$N_1(t)$	$N_3(t)$

$$N_1(t+1) = 0.4 N_0(t) \quad N_0(t+1) = 2 N_1(t) + 1.5 N_2(t)$$

$$N_2(t+1) = 0.3 N_1(t)$$

$$N_3(t+1) = 0.1 N_2(t)$$

$$N_0(t+1) = 0N_0(t) + 2N_1(t) + 1.5N_2(t) + 0N_3(t)$$

$$N_1(t+1) = 0.4N_0(t)$$

$$N_2(t+1) = 0.3N_1(t)$$

$$N_3(t+1) = 0.1N_2(t)$$

$$N(t+1) = \begin{bmatrix} N_0(t+1) \\ N_1(t+1) \\ N_2(t+1) \\ N_3(t+1) \end{bmatrix} = \begin{bmatrix} 0 & 2 & 1.5 & 0 \\ 0.4 & 0 & 0 & 0 \\ 0 & 0.3 & 0 & 0 \\ 0 & 0 & 0.1 & 0 \end{bmatrix} \begin{bmatrix} N_0(t) \\ N_1(t) \\ N_2(t) \\ N_3(t) \end{bmatrix}$$

Given initial population

$$N(0) = \begin{bmatrix} 1000 \\ 200 \\ 100 \\ 10 \end{bmatrix}$$

$$N(1) = \underbrace{\begin{bmatrix} 0 & 2 & 1.5 & 0 \\ 0.4 & 0 & 0 & 0 \\ 0 & 0.3 & 0 & 0 \\ 0 & 0 & 0.1 & 0 \end{bmatrix}}_{\text{Leslie matrix}} \begin{bmatrix} 1000 \\ 200 \\ 100 \\ 10 \end{bmatrix}$$

Leslie matrix

$$N(1) = \begin{bmatrix} 550 \\ 400 \\ 60 \\ 10 \end{bmatrix}$$

$$N(2) = \begin{bmatrix} 550 \\ 400 \\ 60 \\ 10 \end{bmatrix}$$

$$N(2) = \begin{bmatrix} 890 \\ 220 \\ 120 \\ 6 \end{bmatrix}$$

$$N(2) = L \cdot L \cdot N(0)$$

$$= L^2 N(0)$$

$$N(3) = L^3 N(0)$$

$$N(t) = L^t N(0)$$